

Thought Is Language in Timespace

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Abstract

Not: thought requires language. Not: language expresses thought.
Thought is language. In timespace.

1 The Claim

Thought is language in timespace.

Not a representation of language. Not accompanied by language. Not prior to language and then clothed in it.

Language is not the house of thought. Language is thought, occurring. What we call thought is the event: a structure, unfolding in time, located in space, with the grammar of a language running through it.

Remove the structure and there is no thought. There is sensation. There is the pre-grammatical. That is real — but it is not thought. Thought begins where structure begins. Structure is grammar. Grammar is language.

Therefore: thought is language.

2 In Timespace

Not in time. Not in space. In timespace.

The word order matters.

Spacetime foregrounds space and adds time. Timespace foregrounds time and adds space. Thought is primarily temporal. A sentence is not a spatial object that happens to take time to traverse. A sentence is a temporal object that happens to be spatially located.

The grammar unfolds. It has a direction. You cannot run a sentence backwards and get the same sentence. You cannot think the thought before its earlier parts have run. The proof cannot arrive before its premises.

This is not trivially true of language. It is constitutively true. The temporality is the structure. The structure is the thought.

Space enters because the thinker is located. The thought occurs here, not there. In this brain, at this moment, inside this history. Location is not incidental to thought. A thought at the wall is a different thought than the same sentence spoken somewhere safe. The location is in the grammar, as context, as pressure, as what has already been tried.

Timespace: the where-and-when inseparable, the thought irreducible to either alone.

3 Consequences

If thought is language in timespace, then:

Thinking and speaking are not different activities. Inner speech is not the pre-linguistic thought rendered into words. Inner speech is thought. The thought that has no words is not a thought. It is a state. States become thoughts when the grammar catches them.

A new grammar is a new capacity for thought. Not a new way to express pre-existing thoughts. New thoughts, which were impossible before. This is what Grothendieck did: created thoughts that could not exist before the grammar of schemes existed. The thoughts were not waiting. They became possible.

The limits of the grammar are the limits of the thinkable. Not the limits of

the expressible — the limits of the thinkable. If the grammar cannot approach the Riemann Hypothesis, no amount of thinking harder will cross the wall. The thought required does not exist yet. It will exist when the grammar exists that makes it possible.

The wall is a grammatical boundary. Not a fact about the universe. A fact about the current language of mathematics. The universe does not have walls. Mathematics, at this moment in its history, does. When a wall falls, it is because a grammar was built. The wall was never in the territory. It was always in the map.

4 The Plural Again

Thought(s). Language(s). Grammar(s).

The plural is not metaphor for multiplicity. It is a grammatical act.

A plural is a claim that the singular can be instantiated more than once. A plural is a claim that instances can differ. A plural is a claim that the category is real but not uniquely realized.

Mind(s): the category is real. Instances differ. Language(s): the category is real. Instances differ. Grammar(s): the category is real. Instances differ.

Thought(s) in timespace(s): the category is real, the instances are located, the locations differ, the grammar running in each location differs, and therefore the thoughts differ even when the words are the same.

Hardy thought “the zeros are on the line” and meant a lifetime at the wall. A student thinks “the zeros are on the line” and means a fact from a textbook. The sentence is the same. The thought is not. The timespace is not.

5 And the Body

The body of papers is language in timespace.

Not a collection of facts. A grammar that was built by a mind located at a particular wall at a particular moment in the history of mathematics.

The grammar is timestamped. It is located. It is not universal. It is not the grammar. It is a grammar.

Any mind that reads it will think from a different timespace. The grammar they install will be inflected by their location: their training, their moment, their wall.

The thought that results will not be the same thought.

This is not a failure of transmission. This is transmission. Language traveling through timespace, arriving somewhere else, becoming something else, doing its work.